

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Enhancing Emergency Department Triage with Artificial Intelligence

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Student GITHUB: <https://github.com/imaniiz/CariSurg-Portfolio>

Statement of the Problem:

Mercer General Hospital's nurse-led emergency department manages between 70 and 95 patients daily, with volumes exceeding 150 patients during surge periods, placing significant pressure on triage decisions and patient prioritisation. Recent studies have shown that AI-assisted triage models can achieve classification accuracies of up to 80% and improve patient prioritisation (1–3). However, these models have largely been developed using single-site, non-Caribbean datasets and have not been evaluated within Caribbean emergency department workflows. We propose a 12-week pilot to assess whether an AI-assisted decision-support system can improve triage consistency using de-identified Mercer emergency department data.

Previous Work:

Research Paper 1: Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review (1)

This research paper addresses the knowledge gap pertaining to the use of artificial intelligence (AI) and machine learning (ML) in emergency department triage and examines how they affect clinical outcomes. The authors conducted a scoping review to examine the existing use cases of AI in triage by performing a systemized search of the EMBASE, Ovid MEDLINE and Web of Science databases which identified 29 studies published between 2013 and 2023. Overall, the findings showed that ML models consistently outperformed in risk assessment, disease identification, resource allocation, quality of patient care and hospitalization prediction compared to traditional triage methods. While no single AI model was universally superior across all applications, Random Forest, XGBoost and Gradient Boosting models consistently demonstrated the highest predictive performance. The review also considered the ethical issues relating to data and FDA approvals and was limited by its search criteria which excluded papers based on quality appraisal, language and region limiting its overall comprehensiveness and generalizability.

Research Paper 2: TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers (2)

Emergency department overcrowding, increasing case complexity and aging populations have put a strain on healthcare resources which warrants the need for more efficient triage systems. This study aims to explore the integration of AI in triaging systems by evaluating several AI models

using data from 1267 patients admitted to two emergency departments from October 2016 to September 2017. Their predictions, based on the Korean Triage Acuity Scale (KTAS) were compared against assessments made by three triage experts. The results showed that the Support Vector Machine (SVM) and Gradient Boosting Machine (GBM) models exhibited the highest prediction accuracies of 78.7% and 79% respectively while a stacked model combining SVM, GBM and Logistic Regression (LR) produces the best overall performance. A key limitation of this study was its reliance on a single dataset which may limit its generalizability across all clinical contexts.

Research Paper 3: Predicting Emergency Severity Index (ESI) level, hospital admission and admitting ward in an emergency department using data-driven machine learning(3)

Emergency departments often face challenges in accurately prioritising patients and allocating resources due to increasing patient volumes and case complexity. This research paper aims to address this problem by developing the XGBoost machine learning model using data from 653,546 emergency department visits collected over 6 years at Mater Dei Hospital in Malta to predict Emergency Severity Index (ESI) level, hospital admission and admitting ward. The model achieved accuracies of approximately 76% for ESI prediction, 83% for admission prediction and 86% for ward prediction, demonstrating the potential of machine learning to support triage and operational decision-making. The study was however limited by its use of a single hospital dataset, limiting the generalisability of the findings to other patient populations.

Research Paper 4: AI-driven Triage in Emergency Departments: A Review of Benefits, Challenges and Future Directions (4)

Traditional triage systems in emergency departments face increasing challenges due to overcrowding, inconsistent patient prioritisation and resource constraints. This study analysed research papers published between 2015 and 2024 sourced through PubMed, Scopus, IEEE Xplore and Google Scholar in order to examine the benefits and limitations of AI-driven triage systems. The findings suggest that AI-driven triage systems can improve patient prioritisation through consistency and objectivity, support real-time decision making through analytics, enhance resource allocation and are adaptable in mass casualty events. The study also identified future opportunities such as algorithm refinement for all clinical contexts, integration with wearable technology and policy development. However, the review is limited by its reliance on existing literature and its limited study selection criteria which may affect its generalizability.

Research Paper 5: Qualitative Exploration of Patient Flow in a Caribbean Emergency Department(5)

This study investigated the organisational factors affecting patient flow in a Caribbean emergency department, addressing the need to better understand barriers and facilitators within developing healthcare systems. The researchers used a qualitative approach involving non-patient observations, observational process mapping and informal interview collecting 203 hours of observations and mapping 143 patient journeys in a major emergency department in Trinidad and Tobago. The study found that patient flow was influenced by six key factors: organisational work processes, department design and layout, material resources, nursing staff levels and roles, non-clinical staff and external departments, with nursing shortages and lack of inpatient beds identified as major barriers. The authors also noted that the study conducted that its use of a single site with one observer limited its transferability and generalisability of the findings to other healthcare settings.

Research Paper 6: A Human-Centered Hybrid AI Framework for Optimising Emergency Triage in Resource-Constrained Settings (6)

Traditional emergency triage systems are often affected by clinician workload, cognitive biases and resource limitations, which can lead to inconsistent patient prioritisation and delayed treatment. This study proposed a human-centred hybrid AI framework that combines cognitive modelling, explainable machine learning, resource-aware optimisation and a human-in-the-loop interface to support emergency triage decision-making. Through simulated and pilot implementations, the framework achieved approximately 90% triage accuracy, reduced false negatives from 12% to 5%, decreased clinician decision-making time by 31%, and improved critical resource utilisation by 20%. The system also allowed clinicians to override AI recommendations and provide feedback, supporting transparency and continuous learning. A key limitation identified by the authors was the small dataset used for development and testing, which may limit the generalisability of the findings across different healthcare settings.

Research Paper 7: LoRaWAN for Smart Medical Emergency Service (EMS) Integrating with AI(7)

This paper addresses delays in emergency medical response, particularly in rural and low-connectivity areas where patients may not receive timely care during critical emergencies. The authors conducted a survey of existing research on LoRaWAN-enabled healthcare systems and proposed a conceptual framework called SmartMed-LoRaWAN, which combines wearable health sensors, LoRaWAN communication, AI-driven anomaly detection, patient triage and telemedicine support. The proposed framework aims to improve response times and patient outcomes by enabling continuous patient monitoring, emergency detection, automated alerts and remote medical guidance. The study identified several challenges including limited bandwidth, data security concerns and scalability issues when supporting multiple patients simultaneously. Since the system has not yet undergone large-scale clinical validation, its real-world effectiveness therefore remains uncertain.

Research Paper 8: Data Sharing Barriers and Enablers for Healthcare Artificial Intelligence (AI) in Low and Middle-Income Countries (8)

The implementation of artificial intelligence in healthcare within low and middle-income countries is often hindered by challenges related to data availability and sharing. This systematic review aimed to identify the barriers and enablers affecting healthcare data sharing for AI applications in these settings by conducting a literature search across PubMed, SCOPUS, Embase, Web of Science and the Association for Computing Machinery (ACM) databases. Of the 2,471 records identified, 22 studies from 13 countries met the inclusion criteria. The findings revealed that the most significant barriers were technical limitations, such as inadequate hospital infrastructure, and data-related challenges, including the lack of interoperability standards. Key enablers included government support, clear regulatory policies and the transition from paper-based to electronic data collection systems. The study concluded that addressing these barriers could support context-specific AI development in resource-constrained healthcare systems; however, the review was limited by the relatively small number of eligible studies available from low- and middle-income countries.

Research Paper 9: Adoption of Artificial Intelligence in Healthcare: A Survey of Health System Priorities, Successes and Challenges(9)

Healthcare systems face increasing challenges related to clinician burnout, operational inefficiencies and patient safety, creating interest in the adoption of artificial intelligence (AI) tools. This study evaluated the state of AI adoption across 43 United States health systems through a cross-sectional survey conducted in Fall 2024, examining the deployment and perceived success of 37 AI use cases across 10 categories. The results showed that AI adoption varied considerably, with generative AI tools for clinical documentation demonstrating the highest levels of implementation and success, while imaging, radiology and clinical risk stratification tools showed more moderate success despite widespread deployment. The most significant barriers to AI adoption were identified as immature AI tools (77%), financial concerns (47%) and regulatory uncertainty (40%). The authors concluded that while AI has considerable potential to improve healthcare delivery, broader adoption will require stronger evaluation frameworks, governance structures and implementation strategies. A limitation of the study is that it surveyed only 43 large U.S. health systems, meaning the findings may not be representative of smaller healthcare organizations or healthcare systems in other countries.

Research Paper 10: How Explainable Artificial Intelligence Can Increase or Decrease Clinicians' Trust in AI Applications in Health Care: System Review (10)

Artificial intelligence has the potential to improve clinical decision-making, but clinicians may be reluctant to use AI systems when they do not understand how recommendations are generated. This systematic review investigated the impact of explainable artificial intelligence (XAI) on clinicians' trust in AI-driven healthcare applications by reviewing studies identified through searches of the PubMed and Web of Science databases. Of the 778 articles screened, 10 studies met the inclusion criteria and were analysed. The review found that five studies reported increased clinician trust when explanations were clear, concise and clinically relevant, while other studies showed that explanations could either increase or decrease trust depending on their complexity and quality. The authors concluded that explainability can improve AI adoption, but excessive trust in AI recommendations may be just as harmful as distrust, highlighting the importance of achieving an appropriate balance between human judgement and AI support. A key limitation of the review was the small number of eligible studies and the moderate to high risk of bias present across all included studies, limiting the generalisability of the findings.

Proposed Solution:

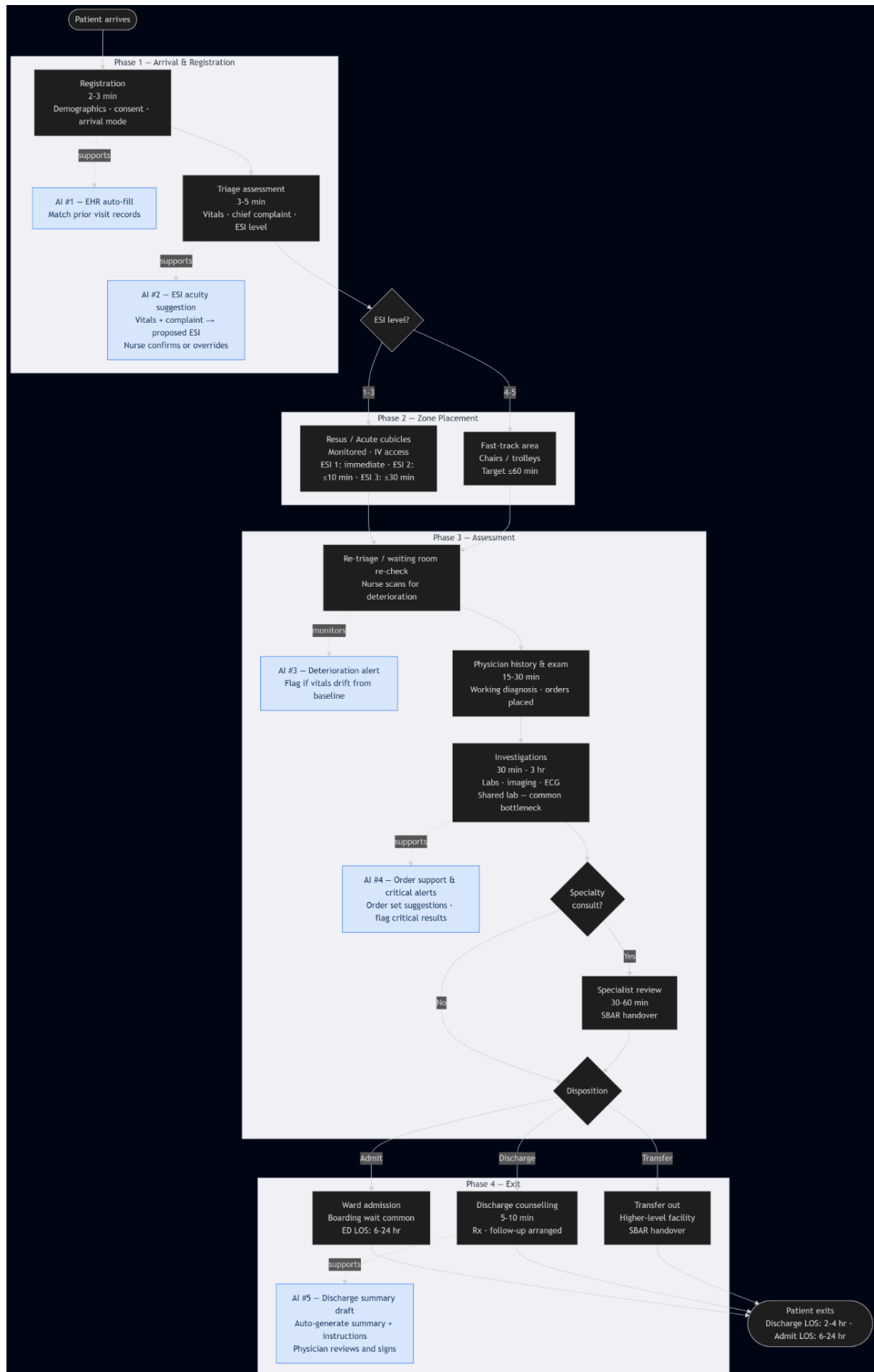
This project proposes the development and evaluation of an AI-assisted triage decision-support system for Mercer General Hospital's Emergency Department. The system is designed to support, rather than replace clinical judgement by providing triage nurses with evidence-based recommendations during the initial patient assessment and subsequent re-triage of waiting patients. Using information already collected during the triage process, including patient demographics, chief complaint, vital signs, pain score, medical history and arrival mode, the system will analyse patient data and generate a recommended Emergency Severity Index (ESI) category along with a confidence score and explanation for recommendation. The system may also identify patients at increased risk of deterioration while waiting for physician assessment, allowing earlier escalation when required.

The proposed solution is intended to fit within existing nurse-led triage workflow and operate within the limited 3–5 minute triage assessment window. AI recommendations would be presented as clinical decision support only, with the triage nurse retaining full responsibility for the final triage decision. This approach aims to improve consistency in patient prioritisation while maintaining clinician oversight and trust. The following areas of the current Mercer General Emergency Department's Workflow were prioritised for AI plugins:

1. Registration
2. ESI acuity suggestion at triage
3. Waiting room deterioration alert
4. Investigation ordering and critical result flagging
5. Discharge summary drafting

In addition to supporting triage decisions, the project highlights the importance of structured digital data capture. By digitising information currently recorded on paper, the system could facilitate future quality improvement initiatives, operational reporting, performance monitoring and the development of more advanced predictive models. A 12-week pilot study will evaluate the feasibility, predictive performance and workflow compatibility of the proposed system within a Caribbean emergency department setting.

Flow Diagram of Mercer General Workflow:



Constraints and Stakeholders:

Workflow Constraints

The Mercer General ED operates in a resource constrained environment characterised by high patient volumes, limited bed capacity, shared diagnostic services and a predominantly paper-based workflow. Any proposed solution must heavily consider the operational constraints of the ED. The following constraints represent key factors that could influence the successful adoption and effectiveness of the proposed system:

1. Constraint #1 - Paper-based documentation and fragmented information flow

Mercer General's ED captures triage decisions mainly on the first page of a paper intake record. Subsequent clinical pages which aim to collect other valuable patient medical data and history are less consistently completed. Information can also be lost during verbal handoffs, re-triage, shift changes and specialist review. This affects an AI system because the relevant clinical details may not be available at the moment of triage.

2. Constraint #2 - Shared diagnostic services creating investigation bottlenecks

Mercer General's ED shares a laboratory, radiology and pharmacy services with the rest of the hospital. As a result, ED investigations compete with inpatient and elective hospital demand, causing lab and imaging turnaround times of 30 minutes to 3 hours. This limits an AI system because a triage model cannot rely on test results that are not yet available.

3. Constraint #3 - Boarding and limited physical space blocking patient flow

Mercer has around twelve acute cubicles, two resuscitation bays, four short-stay observation beds, six fast-track spaces and a waiting room of about 20 seats. When admitted patients remain in the ED for 2 - 24 hours waiting for ward beds, they occupy cubicles or observation spaces needed for incoming patients. This creates crowding at the zone placement stage and can delay care for the newly triaged ESI 2-3 patients. The AI system should therefore help prioritise escalation and determine which patients are waiting on what ED services.

Stakeholders

The successful implementation of an AI-assisted triage system depends not only on the technical performance but also the needs and acceptance of the people who interact with it. Understanding the perspectives of the stakeholders within the Mercer General Emergency Department and their distinct responsibilities, properties and concerns are essential to ensuring the proposed system effectively supports existing workflows.

Table 1: Mercer General Emergency Department Stakeholders

Mercer General Stakeholder	Primary Concern
Patients and Families	Receiving timely and safe care while receiving clear communication during long wait times
Triage Nurses	Ensuring accurate ESI assignment without increasing workload or missing a deteriorating patient in the waiting room
ED Physicians and Consultants	Receiving accurate clinical information that supports decision-making without creating unnecessary alerts or additional work
Hospital Administration and Policymakers	Ensuring patient safety, regulatory compliance, efficient resource utilisation and measurable improvements in ED performance
Data and IT Infrastructure Team	Maintaining reliable data capture, privacy, security and integration with existing workflows

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